Q.No:10

DS	Session	Arrays	Question Information	Level 1	Challenge 30

Problem Description:
Bear Grylls is a forest lover, so he spends some free time taking care of many of her loved ones' animals. He likes to offer them treats, but wants to do that in an impartial way.

Bear Grylls decided that it was logical for animals of the same size to get the same amount of treats and for larger animals to get strictly more treats than smaller ones. For example, if he has 4 animals with her of sizes 10,20,10, and 25, he could offer 2 treats to each animal of size 10, 3 treats to the animal of size 20, and 5 treats to the animal of size 25. This requires her to buy a total of 2+3+2+5=12 treats. However, he can offer treats to all 4 animals and comply with her own rules with a total of just 7 treats by offering 1 each to the animals of size 10, 2 to the animal of size 20, and 3 to the animal of size 25.

Help Bear Grylls plan her next animal day. Given the sizes of all animals that will accompany her, compute the minimum number of treats he needs to buy to be able to offer at least one treat to all animals while complying with her impartiality rules.

Constraints:
$1 \le T \le 100$.
$1 \le Si \le 100$, for all i.
$2 \le N \le 100$.

Input Format:
The first line of the input gives the number of test cases, T. T test cases follow.

Each test case consists of two lines.

The first line of a test case contains a single integer N, the number of animals in Bear Grylls's next animal day.

The second line of a test case contains N integers S1,S2,...,SN, representing the sizes of each animal.

Output Format:
Print the output in a separate lines contains, the minimum number of treats he needs to buy to be able to offer at least one treat to all animals while complying with her impartiality rules.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
4
10 20 10 25
5
7 7 7 7 7
2
100 1
```

EXPECTED OUTPUT

```
7
5
3
```

Test Case 2

INPUT (STDIN)

```
5
5
8 6 4 8 9
3
8 5 2
4
101 120 110 215
5
1711 171 711 71 71
2
110 11
```

EXPECTED OUTPUT

```
13
6
10
11
3
```

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
`int s[MAXN];`	`void sol()`	`read(s[i])`

Code:

```c
#include <stdio.h>

#define MAXN 100
#define read(x) scanf("%d", &(x))

int s[MAXN];

void sol()
{
int n;
int i, j;
int temp;
int treats = 0;
int currentTreat = 1;
```

```
printf("Input value:\n");

read(n);

for(i = 0; i < n; i++)
{
read(s[i]);
}
for(i = 0; i < n - 1; i++)
{
for(j = i + 1; j < n; j++)
{
if(s[i] > s[j])
{
temp = s[i];
s[i] = s[j];
s[j] = temp;
}
}
}
treats = 0;
currentTreat = 1;

for(i = 0; i < n; i++)
{
if(i > 0 && s[i] != s[i - 1])
{
currentTreat++;
}

treats = treats + currentTreat;
}

printf("\nOutput:\n");
printf("%d\n", treats);
}

int main()
{
sol();

return 0;
}
```

Output:

Test _case 1:

```
Input value:
3
4
10 20 10 25
5
7 7 7 7 7
2
100 1

Output:
7
5
3
```

Test _case 2:

```
Input value:
5
5
8 6 4 8 9
3
8 5 2
4
101 120 110 215
5
1711 171 711 71 71
2
110 11

Output:
13
6
10
11
3
```